

Multiple Choice (10 marks)

1. A reaction that includes energy as one of its reactants is called a(n)
 - (a) endergonic reaction.
 - (b) catabolic reaction.
 - (c) redox reaction.
 - (d) anabolic reaction.
 - (e) exergonic reaction.
2. Explain why simple diffusion is not sufficient to account for the passage of all of the dissolved minerals into the cells of water plants.
 - (a) Simple diffusion can only transport minerals when they are bound to oxygen molecules
 - (b) Simple diffusion cannot occur against a concentration gradient
 - (c) Simple diffusion requires specific carrier proteins for each mineral
 - (d) Simple diffusion only works for organic compounds, not inorganic minerals
 - (e) Simple diffusion can transport minerals in any concentration
3. Which of the following is a second messenger that stimulates release of calcium ions into the cytoplasm?
 - (a) Adenosine triphosphate
 - (b) Nitric oxide
 - (c) Guanosine triphosphate
 - (d) Cyclic GMP
 - (e) Prostaglandins
4. All of the following statements are true EXCEPT
 - (a) insulin decreases storage of glycogen
 - (b) growth hormone stimulates cell division and protein synthesis
 - (c) epinephrine increases blood sugar levels and heart rate
 - (d) thyroxine increases the rate of metabolism
 - (e) vasopressin stimulates water reabsorption in the kidney
5. One group of compounds that may become quite important in birth control is the prostaglandins. What are the advantages of prostaglandins in birth control?

- (a) Prostaglandins are used to enhance mood and have no effect on birth control.
- (b) Prostaglandins increase fertility by promoting ovulation.
- (c) Prostaglandins are able to prevent pregnancy by altering the DNA of the sperm and egg.
- (d) Prostaglandins are known to mediate hormonal action
- (e) Prostaglandins are non-hormonal, reducing the risk of hormonal side effects.

True / False (10 marks)

Answer True or False

- 6. True or False: Crossing-over during meiosis allows scientists to determine the exact sequence of DNA bases in an organism's genome.
- 7. True or False: All protozoans share a uniform reproductive method, resulting in consistent offspring across species.
- 8. True or False: Measuring the physical characteristics of a population is sufficient to assess its genetic equilibrium status.
- 9. True or False: Cyanobacteria are prokaryotic organisms that lack a true nucleus.
- 10. True or False: The vegetal pole does not undergo cleavage more readily than the animal pole in telolecithal eggs.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the various methods used to form recombinant DNA, and explain why translation does not qualify as one of these methods.
- 12. Discuss the factors that lead to the movement of electrons between orbitals in an atom, focusing on how their energy remains constant while their position changes.

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